

ORGAN SYSTEM ANATOMY (MID & POST LEC)

Lecture #2: Nervous System | OPHT 1-1 | PRELIMS

By Accel Jimenez

NERVOUS SYSTEM

★ Nervous System

- Acts as the control center of the body.
- Monitors, coordinates and regulates all our movements, sensation, and internal functions.
- Without a nervous system, we will not feel our environment.
- **Neurons = messenger units.** Sends info through the spinal cord to the muscles and nerves.

Functions of the Nervous System:

- ★ **Sensory Input:** Detects internal and external stimuli
- ★ **Integration:** interprets and stores information, makes decisions (short term or long term memory)
- ★ **Motor Output:** Responds by activating effectors (muscles glands)
- ★ **Homeostasis:** Regulates body processes via rapid communication (when body temp rises, nervous system orders the sweat glands)
- ★ **Mental Activity:** Involves consciousness, memory, emotions, thinking

DIVISIONS OF THE NERVOUS SYSTEM

★ Structural (Parts)

- **Central Nervous System (CNS):** *Brain + Spinal Cord main*
 - command center
 - this where all the info goes through for processing.
- **Peripheral Nervous System (PNS):** *Cranial nerves + Spinal Nerves + Ganglia*
 - Messenger system
 - Personal assistant of CNS.
 - The ones who carry the info to the CNS, and back to the muscles and glands (delivery boys).

★ Functional

- **Somatic Nervous System (SNS):** Voluntary skeletal muscle control
 - Walking, raising of hand
- **Autonomic Nervous System (ANS):** Involuntary (sympathetic & parasympathetic)

- Breathing, beating of heart, digestion, sweating.

- **Enteric Nervous System (ENS):** Specialized network regulating GI tract

- Can work semi-independently (can control digestion even without too much input from nervous system)
- 2nd brain of the gut

CELLS OF NERVOUS TISSUE

- ★ **Neurons:** Excitable cells that transmit/generate electrical signals
- ★ **Glial Cells (neuroglia):** Supporting cells that protect, nourish, and insulate neurons. (caretakers of neurons)
 - Glial cells that makes myelin sheath = makes the signal be faster transmitted

NEURON STRUCTURE

- ★ **Cell Body (Soma):** Contains nucleus, nissl bodies, cytoplasm
 - Nissl bodies: rough endoplasmic reticulum (where proteins of neurons are made) (control center of neuron)
- ★ **Dendrites:** Receive input, carry toward soma. (antennas)
- ★ **Axon:** Conducts impulses away from soma
- ★ **Axon terminals**
 - Axon Hillock - initial segment - trigger zone (where nerve signals generated)
 - Axon terminals

Dendrites (**receive**) - Soma (**decides**) - Axon (**Send info to**) - Axon Terminals (**Act**) - Axon Hillock

TYPES OF NEURONS

★ Structural Neurons

- **Multipolar** (most common, motor neurons)
 - 1 axon, many dendrites
 - Purpose: For motor movement

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- **Bipolar** (retina, olfactory epithelium)
 - Can be found on eyes (retina) and nose (inner part olfactory)
 - Rare, 1 dendrite, 1 axon.
 - Purpose: For special senses
- **Unipolar** (pseudounipolar neurons)
 - Usually found sensory neurons in dorsal root ganglia.
 - Purpose: Transmits sensory information.

★ Functional Classification

- **Sensory (afferent) neurons**
 - Towards the brain
 - Brings info to CNS
- **Motor (efferent) neurons**
 - Responsible for movements, motor controls
 - Away from CNS
- **Interneurons** (association neurons, WITHIN CNS)
 - Afferent and efferent will not be connected without interneurons.
 - Connects Afferent and Efferent neurons.

NEUROGLIAL CELL'S TYPES & FUNCTIONS

Central Nervous System

★ Astrocytes

- Most abundant glial cells in CNS
- Gives structural support of neurons, regulates environment
- Blood brain barrier/ Filter system

★ Ependymal Cells

- Help produce, circulate and monitor cerebrospinal fluid (CSF)

★ Oligodendrocytes

- Responsible for forming myelene sheath in CNS
- Can wrap many axons = production of impulses will be faster.
- Slow to regenerate and sometimes cannot be regenerated after injury.

★ Microglia

- Immune defense cells of CNS
- Phagocytes
- Clears unwanted debris and dead cells, pathogens

Peripheral Nervous System

★ Satellite Cells

- Surround cell bodies in ganglia
- Regulate neurotransmitter level

★ Schwann Cells

- Forms myelene sheets in PNS
- Can wrap axons one at a time
- Help nerve regeneration of neurons after injury.

NERVOUS TISSUE RESPONSE TO INJURY

★ **CNS Neurons:** Limited regeneration (due to inhibitory environment, lack of neurolemma) (protective covering that helps the regeneration)

★ **PNS Neurons:** Potential regeneration if soma intact & schwann cells from regeneration tube

- Peripheral nerve conditions
 - **Saturday-night palsy** - lying on your arms laterally = compression = temporary weakness on that arm
 - **Carpal Tunnel syndrome** - compression on the median nerve on the wrist, weird feeling, but after compression out, ok

★ **Glial Scar formation:** Astrocytes wall of injury site.

- Double edged sword. It isolates the injured area, but also shuts off the nutrients resulting in stopping of regeneration.

MYELINATED V.S UNMYELINATED AXONS

★ Myelinated

- Insulating sheath (oligodendrocytes in CNS, Schwann cells in PNS)
- Saltatory conduction (jumps from node to node = faster)
- Myelinated axon = function is for motor.

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★ Unmyelinated

- No sheath, conduction is continuous and slower.
- Unmyelinated axon = function is for the regulation/digestion of organs. Process before action.

Corpus Callosum: Connects the right and left hemisphere of the brain (white matter outside).

Action Potentials (starts in axon hillock)

- ★ **All-or-none response**, generated at axon hillock (hit it or miss it)

Dendrite - soma - axon hillock - axon - axon terminal

BRAIN

★ Grey Matter (unmyelinated)

- Grey = content is neuronal cell bodies, dendrite, and unmyelinated axons (no fatty myelins)
- Rich in blood supply compared to the inner part.

Outer parts (example is grey matter) are rich in blood supply compared to the inner part.

★ White matter (myelinated)

- White = contains myelinated axons only.

PHASES

1. Resting State (RMP maintained)

- a. Resting membrane potential of neurons: -70 millivolts
- b. Cells have more negative charge inside the cell compared to the outside.
- c. 3 sodium out, 2 potassium in (so it'll be ready)

2. Depolarization

- a. Stimulus reached

3. Repolarization

4. Hyperpolarization

1. Threshold Stimulus is reached (-55 mV)
2. Voltage Gated Sodium channels open initiating depolarization
3. Repolarization begins
4. Hyperpolarization occurs
 - a. Hyperpolarization = membrane voltage is below RMP (-70 mV)

WHITE MATTER	GREY MATTER
White color	Grey color
Located deep in the brain; outer portion of the spinal cord	Located outer layer of brain (cortex); central portion of spinal cord
Composed mainly of myelinated axons and oligodendrocytes	Composed mainly of neuron cell bodies, dendrites, and unmyelinated axons
Function: transmit signals between diff. Brain regions	Function: process and analyze information
Fast signal transmission	Slower signal transmission
Shows structural changes with learning and experience	Undergoes pruning and reorganization during development
Appears bright on T1-weighted MRI images	Appears dark on T1-weighted MRI images